

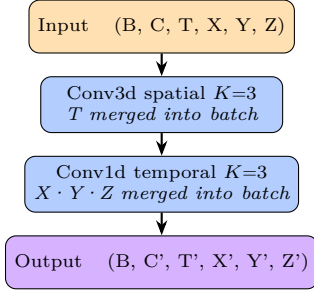
Patchify architecture — PatchEmbed3DPlus1D

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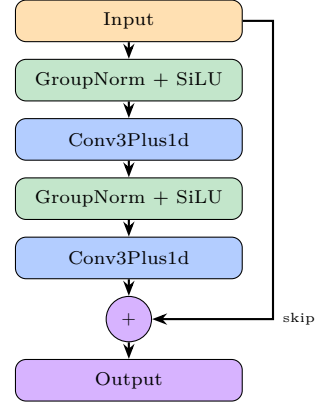
Hierarchical 3D+1D patchify encoder, inspired by the MovieGen TAE TemporalEncoder (tae.py:1052).

Building blocks

Conv3Plus1d — separable substitute for the missing PyTorch Conv4d. Splits a 4D conv into a 3D spatial conv (per frame) followed by a 1D temporal conv (per voxel-position).

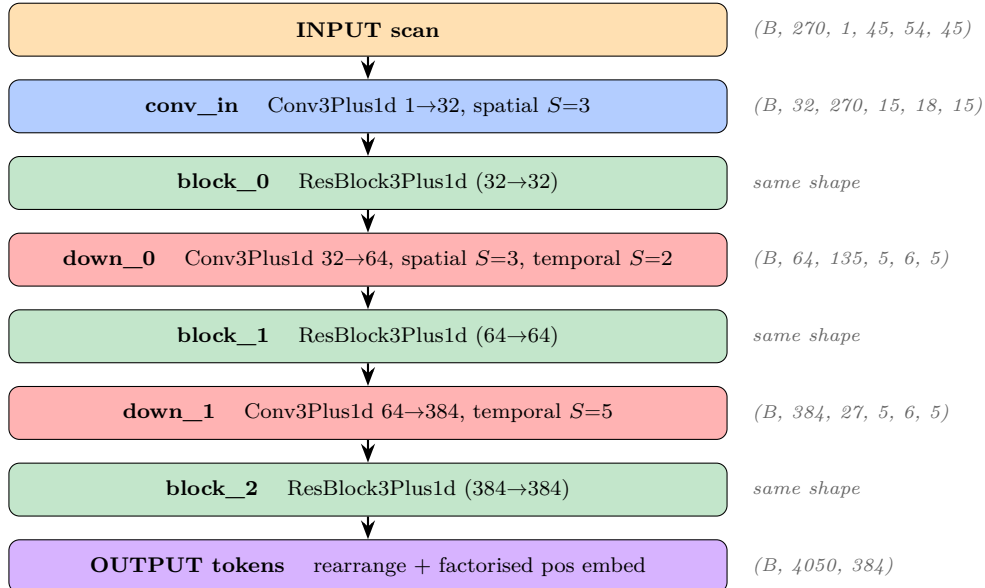


ResBlock3Plus1d — residual block of 2 Conv3Plus1d with GroupNorm + SiLU pre-activation, output added to a skip connection. Same shape in / out.



Full encoder (current multi-source setup, $T = 270$, $\text{temporal_kernel} = 10$)

The corpus is resampled online to $TR = 0.72$ s and cropped to $T_{\text{fixed}} = 270$ frames. *temporal_kernel* is a config knob (the earlier HCP-YA-only setup used $T = 1200$, *temporal_kernel* = 20); with *temporal_kernel* = 10 the total temporal downsample is $\times 10$ and $T_{\text{eff}} = 270/10 = 27$. Note *down_1* temporal stride = *temporal_kernel* / 2.



Parameter breakdown

Layer	What it contains	Params
conv_in	1 Conv3Plus1d (1 $\rightarrow$ 32)	4 000
block_0	2 Conv3Plus1d (32 $\rightarrow$ 32) + 2 GroupNorm	61 696
down_0	1 Conv3Plus1d (32 $\rightarrow$ 64)	67 712
block_1	2 Conv3Plus1d (64 $\rightarrow$ 64) + 2 GroupNorm	246 272
down_1	1 Conv3Plus1d (64 $\rightarrow$ 384), $K_t=5$	1 401 600
block_2	2 Conv3Plus1d (384 $\rightarrow$ 384) + 2 GroupNorm	8 850 432
PositionEmbedding3D	pos_temporal + pos_spatial + pos_cls	68 352
TOTAL		10 700 064

Token grid output : $T_{\text{eff}} = 27$, $N_{\text{spatial}} = 5 \cdot 6 \cdot 5 = 150$, total **4 050 tokens** of dim 384.

Detailed parameter calculation per layer

Each Conv3Plus1d has 2 internal modules : Conv3d(in_c, out_c, K_s) (spatial) + Conv1d(out_c, out_c, K_t) (temporal). Both convs include a bias term.

Conv3d params = $\text{in}_c \cdot \text{out}_c \cdot K_s^3 + \text{out}_c$ **Conv1d** params = $\text{out}_c \cdot \text{out}_c \cdot K_t + \text{out}_c$ **GroupNorm** params = $2 \cdot C$ (scale + shift, both of length C)

1. conv_in = Conv3Plus1d(1 $\rightarrow$ 32, $K_s=3$, $K_t=3$)

spatial	Conv3d(1, 32, k=3)	:	$1 \times 32 \times 3^3 + 32$	=	896
temporal	Conv1d(32, 32, k=3)	:	$32 \times 32 \times 3 + 32$	=	3 104
					4 000

2. block_0 = ResBlock3Plus1d(32) = 2 $\times$ Conv3Plus1d(32 $\rightarrow$ 32) + 2 $\times$ GroupNorm(8, 32)

GroupNorm $\times$ 2	:	$2 \times (2 \times 32)$	=	128
Conv3Plus1d $\times$ 2 :				
spatial each	:	$32 \times 32 \times 3^3 + 32$	=	27 680 each
temporal each	:	$32 \times 32 \times 3 + 32$	=	3 104 each
total one	:	$27 680 + 3 104$	=	30 784
$\times$ 2	:		=	61 568
				61 696

3. down_0 = Conv3Plus1d(32 $\rightarrow$ 64, $K_s=3$, $K_t=3$)

spatial	Conv3d(32, 64, k=3)	:	$32 \times 64 \times 3^3 + 64$	=	55 360
temporal	Conv1d(64, 64, k=3)	:	$64 \times 64 \times 3 + 64$	=	12 352
					67 712

4. block_1 = ResBlock3Plus1d(64) = 2 $\times$ Conv3Plus1d(64 $\rightarrow$ 64) + 2 $\times$ GroupNorm(8, 64)

GroupNorm $\times$ 2	:	$2 \times (2 \times 64)$	=	256
Conv3Plus1d $\times$ 2 :				
spatial each	:	$64 \times 64 \times 3^3 + 64$	=	110 656 each
temporal each	:	$64 \times 64 \times 3 + 64$	=	12 352 each
total one	:	$110 656 + 12 352$	=	123 008
$\times$ 2	:		=	246 016
				246 272

5. **down_1 = Conv3Plus1d(64 → 384, $K_s=3$, $K_t=5$)**

($K_t = \text{temporal_kernel} / 2 = 5$ in the current config; it was 10 in the HCP-YA $T=1200$ setup.)

spatial Conv3d(64, 384, k=3) : $64 \times 384 \times 3^3 + 384 = 663\ 936$
temporal Conv1d(384,384, k=5) : $384 \times 384 \times 5 + 384 = 737\ 664$

1 401 600

6. **block_2 = ResBlock3Plus1d(384) = $2 \times \text{Conv3Plus1d}(384 \rightarrow 384) + 2 \times \text{GroupNorm}(8, 384)$**

The dominant layer — 83% of the total (this is the block point 2 proposes to remove).

GroupNorm $\times 2$: $2 \times (2 \times 384)$ = 1 536

Conv3Plus1d $\times 2$:

 spatial each : $384 \times 384 \times 3^3 + 384 = 3\ 981\ 696$ each

 temporal each : $384 \times 384 \times 3 + 384 = 442\ 752$ each

 total one : $3\ 981\ 696 + 442\ 752 = 4\ 424\ 448$

$\times 2$: = 8 848 896

8 850 432

7. **PositionEmbedding3D (current : $T_{\text{eff}}=27$, $N_{\text{spatial}}=150$, dim =384)**

pos_temporal : 27×384 = 10 368

pos_spatial : 150×384 = 57 600

pos_cls : 1×384 = 384

68 352

Grand total: $4\ 000 + 61\ 696 + 67\ 712 + 246\ 272 + 1\ 401\ 600 + 8\ 850\ 432 + 68\ 352 = 10\ 700\ 064$ params.

Differences vs MovieGen TAE TemporalEncoder

Aspect	MovieGen TAE	Ours (PatchEmbed3DPlus1D)
Domain	Video (T, H, W)	fMRI volumes (T, X, Y, Z)
Building block	Conv2Plus1d (2D spatial + 1D temporal)	Conv3Plus1d (3D spatial + 1D temporal)
Hierarchical levels	4 (default $\text{ch_mult}=(1,2,4,8)$)	3 (channels 32 / 64 / 384)
ResBlocks per level	2	1
Attention in encoder	Yes (at specified resolutions)	None
Middle bottleneck section	2 ResBlocks + 1 AttnBlock	None
Output	Latent tensor $\mathbf{z}$ (for decoder)	Token sequence ($\mathbf{B}$, $T_{\text{eff}} \cdot N_{\text{spatial}}$, 384) for ViT